THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING: LEVEL 1 EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

ENGINEERING MECHANICS AND MATERIALS – CIVL1013

DATE: 18 May, 2011 TIME: 9:30 am – 12:30 pm (3 hours)

Answer any 2 questions from Section A and any 2 questions from Section B.
All questions carry equal marks.
Use a separate answer book for Section A and Section B.

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

SECTION A

You are required to answer 2 out of 3 questions in Section A.

Question 1

A simply supported beam is subjected to a central point load of magnitude P (kN) as shown in Figure 1(a). The beam is made from a box section, formed from welding four 25 mm thick steel plates together, as shown in Figure 1(b). The elastic modulus of the steel is 200,000 MPa and the maximum allowable bending stress is 250 MPa.

Determine the following:

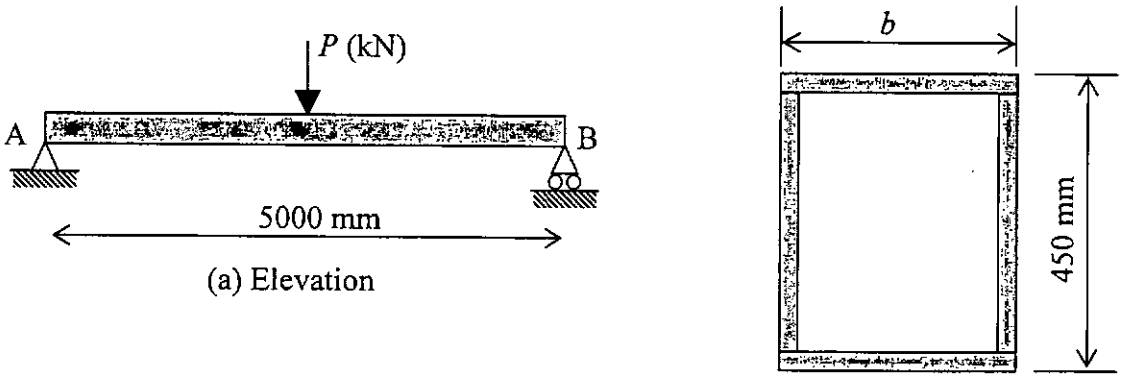
(a) An expression for the second moment of area (I) as a function of the width of the section, b. **(3 marks)**

(b) The width b of the box section required to support a load P of 1000 kN (note: neglect shear). **(6 marks)**

(c) What is the curvature of the beam due to the conditions in Part 1(b)? **(3 marks)**

(d) What load P can be carried by the beam if the width of the box section b equals 200 mm (note: neglect shear)? **(6 marks)**

(e) If the width of the box section is 200 mm and the load P is 600 kN, calculate the maximum section shear stress. (7 **marks**)



(a) Elevation

(b) Section

Figure 1.

Question 2

Consider the cantilever beam of constant flexural rigidity EI, shown in Figure 2, which is subjected to a linearly varying distributed load of maximum intensity w_o.

Determine the following:

(a) All reactions. **(4 marks)**

(b) The equation of the deflection curve with respect to distance x from the fixed support, A, using the differential equation method. **(16 marks)**

(c) Equations for the deflection at the support A and the free end B utilising the result from Part 2(b). **(5 marks)**

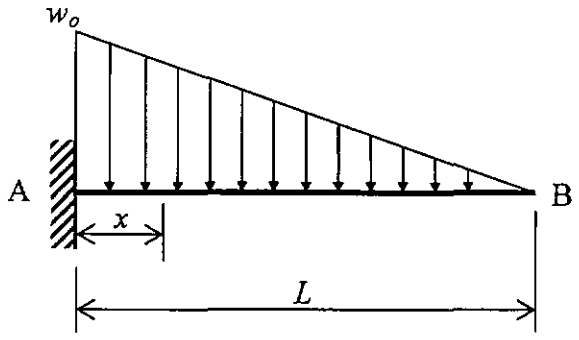
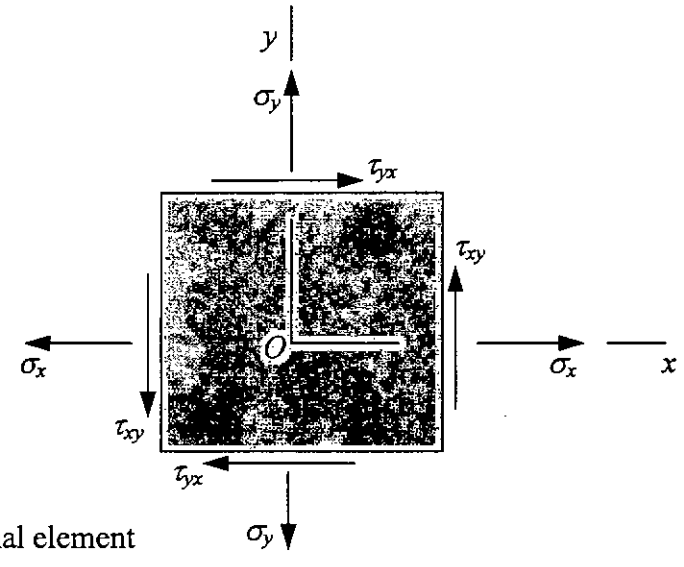


Figure 2.

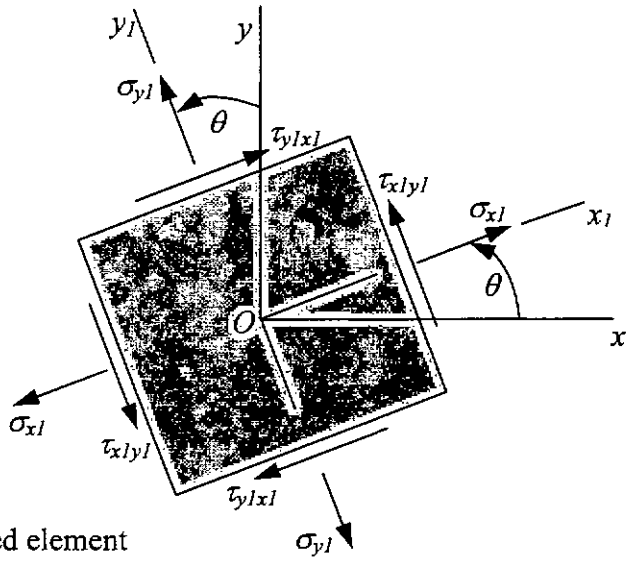
Question 3

Consider the two-dimensional plain-stress elements shown below in Figure 3. Derive the equation for σ_{x1} from first principles which will enable transformation of stress from the original element in Figure 3(a) to the rotated element in Figure 3(b).

(Given $\sec^2\theta = 1 + \tan^2\theta$; $\cos^2\theta = \frac{1}{2}(1 + \cos 2\theta)$; $\sin^2\theta = \frac{1}{2}(1 - \cos 2\theta)$; $2\sin\theta\cos\theta = \sin 2\theta$ } **(25 marks)**



(a) Original element



(b) Rotated element

Figure 3.

- END OF SECTION A -